

Auxiliary Source Frames for Integer Multiplication

Concentrating auxiliary rank into contiguous recursive blocks

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Abstract

The batched recursion of PR #10 makes a contiguous block of t address fields cost one recursive call of width t , so a role's rank is cheaper when it is concentrated in one large edge. Auxiliary roles may start in any fixed frame, provided they end in that frame plus the identity. Starting every stage-two auxiliary role of the interchange network in the frame of its first gate removes its rank- $(h^2 - h)$ entrance edge and turns its exit into one projector of rank $m - h$. The rank sum and deficit are unchanged, but the interchange saving rises from $246/10^9$ to $154/10^8$. A third whole-residual class in the complex network raises its saving to $18/10^7$, so it does not bind. The conclusion is conditional on the pinned upstream multiplication theorem and the retained interfaces of PR #10 and its predecessors.

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{7699}{10^{10}} > 2^{-21}.$$

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1 Result and retained framework

The model is a fixed finite-alphabet Turing machine with a fixed number of one-dimensional tapes. This note changes two finite choices in the construction of PR #10 by icekylinx [9], pinned at commit

62691e395a0458ce089a1c7b5d89e74291e95e29.

That construction consumes the ternary five-subset network and the paired complex producer of Zhihao Chen’s PR #7 [8] unchanged. Sections 2 to 5 reproduce the batched framework of PR #10 verbatim. Section 6 records its exact complex exponent and its 2^{-23} assembly, which the present note supersedes. Sections 7 to 9 contain the new arguments. No scalar circuit, label, producer or analytic interface changes. Only two choices change: the source and sink matrices of the stage-two auxiliary roles, and the set of complex residuals applied as whole children.

2 Large rational projectors

2.1 Batching a large-rank projector edge

The usual matrix-shear compiler uses one smaller interchange per rank unit. For a projector of rank close to the ambient dimension, most of these pivot interchanges can instead be one interchange of two contiguous longer fields. The construction below does not gather separated fields or reverse their digits.

A simultaneous rational change of basis. Let $\mathcal{P}$ be a fixed finite family of rational idempotents of dimension m . For $P \in \mathcal{P}$ with $m/2 < a := \text{rank } P < m$, put $r = m - a$. There is one $S \in \text{GL}_m(\mathbb{Q})$ such that, for every such P , the submatrix of SPS^{-1} on its first r rows and last r columns is invertible.

To prove this, first consider one rank- a idempotent. Every rational idempotent is rationally similar to $\text{diag}(I_a, 0)$, by concatenating rational bases of its image and kernel. Put $t = m - 2r > 0$. On the first and last r coordinates take r disjoint rank-one blocks

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

pairing coordinate i with $m + 1 - i$, and take the identity on the middle t coordinates. This is a rational rank- a idempotent whose indicated corner is invertible. Hence the determinant of that corner of SPS^{-1} is not the zero rational function of the entries of S . Clear powers of $\det S$ from its denominator. The product of the resulting nonzero polynomials for the finite family, and $\det S$, is nonzero over $\mathbb{Q}$. A nonzero polynomial over the infinite field $\mathbb{Q}$ cannot vanish at every rational point. This proves simultaneous existence.

The choice is effective: enumerate integer matrices and retain the first invertible one satisfying all the indicated rational determinant tests. This search terminates. The family and m are fixed, so this finite calculation is part of the fixed-machine preparation. Conjugate every terminal and gate matrix by this same S . Common gate matrices, every edge rank, the scalar circuit, and the endpoint identity are preserved, since $SIS^{-1} = I$. The address prime is then chosen after this conjugation and the triangular factorizations, avoiding their finitely many denominators and invertible diagonal numerators as in the original construction.

Only the edges selected for batching need belong to $\mathcal{P}$. All other matrices can continue to use the original rank-by-rank compiler. In the present finite network, every selected edge is a nonzero scalar multiple of an idempotent: for nested nondegenerate labels $U \subset V$,

$$(P_V - P_U)^2 = P_V - P_U,$$

and reversing the edge only changes its sign. Scalar multiples do not change the pivot locations and are absorbed into the triangular factors.

The contiguous middle pivots. Let $P^2 = P$ have rank $a > m/2$, let $r = m - a > 0$, and assume its first- r -row, last- r -column corner is invertible. Split the ordered coordinates into

$$L = [1, r], \quad M = [r + 1, m - r], \quad R = [m - r + 1, m].$$

The complement $Q = I - P$ has rank r . Write $Q = UV$, where U has r columns, V has r rows, and $VU = I_r$. Such a factorization comes from a basis of $\text{im } Q$. As L and R are disjoint,

$$P_{L,R} = -U_L V_R.$$

Its invertibility makes both square matrices U_L, V_R invertible.

Run the lower-lower elimination in the retained physical order. Its first r pivots occupy the rows L and the columns R , in some permutation. Indeed their corner is invertible, so at each of these steps the next row has a nonzero entry in an as-yet active column of R , which precedes all other candidates in the rightmost-pivot rule. The active Schur complement on rows $M \cup R$ and columns $L \cup M$ has middle block

$$\begin{aligned} P_{M,M} - P_{M,R} P_{L,R}^{-1} P_{L,M} &= I - U_M V_M - (-U_M V_R)(-V_R^{-1} U_L^{-1})(-U_L V_M) \\ &= I_{m-2r}. \end{aligned}$$

Thus the next $m - 2r$ pivots are precisely the diagonal positions (i, i) with $i \in M$. Columns L lie to their left and are cleared by the allowed lower-triangular column operations. The remaining block has rank zero, because the number of isolated pivots is already $r + (m - 2r) = m - r = a$.

Consequently the factorization $P = E_1 \Pi E_2$, with E_1, E_2 invertible lower triangular, has exactly r corner pivots and one contiguous identity block of size

$$t = m - 2r = 2a - m.$$

The same conclusion holds for cP with any fixed nonzero rational c . No assertion that an arbitrary rank- a matrix has this pivot profile is needed; the idempotent identity is what makes the middle block exactly an identity matrix.

One call for the identity block. Subdivide each original contiguous chunk into m consecutive fields of width b . The triangular factors still cost $O(V)$ in the retained later-field affine primitive. Each of the r corner pivots uses one ordinary width- b interchange. For all $i \in M$ together, perform

$$D_i \leftarrow D_i + H_i, \quad (H_M, D_M) \leftarrow (D_M, H_M), \quad D_i \leftarrow H_i - D_i,$$

over $\mathbb{Z}/q^b\mathbb{Z}$. The first and last maps are coordinatewise later-field updates, each using a fixed number of linear scans. The middle map is a single interchange of the two contiguous width- tb chunks H_M, D_M . It preserves the order of their b -digit fields and hence implements all diagonal

pivots exactly. The intervening fields are spectators. In particular, addition is componentwise modulo q^b , not addition with carries across the concatenated width- tb chunks.

The edge cost therefore has r recursive calls of width b and one of width tb , in place of a calls of width b , with the same $O(V)$ nonrecursive work. If $a = m$, one may leave the edge in its original m -singleton form. This avoids any same-width recursion. For every batched edge in this paragraph, $0 < t \leq m - 2$, so every child contracts its width by a fixed factor less than one.

The resulting characteristic expression. Let $\mathcal{B}$ be the selected large-rank projector edges, let $r_e = m - a_e$ and $t_e = m - 2r_e$ for $e \in \mathcal{B}$, and retain the total rank sum s . For an exponent $0 < \tau < 1$, the volume-weighted recursive contribution is

$$\mathcal{C}(\tau) = \frac{1}{W} \left[\left(s - \sum_{e \in \mathcal{B}} t_e \right) m^{-\tau} + \sum_{e \in \mathcal{B}} \left(\frac{t_e}{m} \right)^\tau \right]. \quad (2.1)$$

At $\tau = 1$ this is exactly $s/(Wm)$: batching preserves the existing rank deficit. The strict inequality $\mathcal{C}(\tau) < 1$ is the new time-exponent comparison. It uses only the original rank sum and the count of selected ranks; no complete edge-rank histogram is needed. Non-power widths use equal fields of width $\lfloor e/m \rfloor$, at most $m - 1$ separately handled high digits, and the same fixed-tape row recursion. With sufficient rows the nonrecursive work is $O(V)$, so the usual strong induction gives $O(Ve^\tau)$ whenever (2.1) is strict. The separate row and arbitrary-width construction specifies this rounding step and removes the initial row divisibility restriction.

3 The explicit interchange recurrence

3.1 Three geometric rank classes in the retained ternary network

The five-subset producer of PR #7 is retained. Put

$$h = 28, \quad v = \binom{28}{5} = 98280, \quad m = h^3 = 21952, \quad R = 11840940, \quad N = v^3,$$

$$W = 2v^2(v + R), \quad L = 3v^2 \binom{h}{2} (h - 2), \quad s = Wm - N + 2L.$$

This subsection needs the producer's proved width upper bound and frame schedule; it does not enumerate its internal addition graph again. If the producer uses fewer than R roles, insert dummy auxiliary roles with identity gates at the ordinary low and high frames of every invocation. Do this before identifying the first- and third-stage banks. Every dummy role restores its arbitrary scalar, has no decreasing edge, and contributes exactly m to the final rank sum. Thus the stated width and the rank classes below are realized exactly, including any padding.

Write $a = h^{j-1}$ at tensor stage j . With the notation of the retained frame proof, the early auxiliary frame is

$$D_0 = (B \otimes F) \otimes Q, \quad \dim D_0 = (a - 1)h,$$

and the final auxiliary frame is

$$D_1 = (A \otimes F) \otimes Q, \quad \dim D_1 = ah.$$

The two auxiliary banks give the following edges.

1. Each role joining stages one and three passes from a frame of dimension h to a containing frame of dimension $m - h$. This is a projector edge of rank $m - 2h$. There are $B = v^2 R$ such roles.
2. Each stage-two auxiliary role passes from its final frame of dimension h^2 to the full sink frame of dimension m . Its rank is $m - h^2$, again with $B = v^2 R$ copies.

There is one further large class on the data wires. The incoming physical X and Y dimensions at stage j are a and $a - 1$. Their first relevant frames have dimensions $(a - 1)h + 1$ and $(a - 1)h$, respectively. Hence both entrance edges have rank

$$d_j = (a - 1)(h - 1).$$

The remaining growth of each data role in that stage is $h - 1$. These dimensions hold for the reverse middle-stage schedule as well: its low source copy and its target-line injection give precisely the same physical frames. At stage three,

$$d_3 = (h^2 - 1)(h - 1) = 21141 > m/2,$$

and there are $2N$ copies. The special negative source matrix contributes one rank-one edge on each physical X source, as in PR #7. Consequently the complete data rank sum is

$$2N \sum_{j=1}^3 (d_j + h - 1) + N = 2Nm - N.$$

All remaining auxiliary changes inside an invocation have rank at most h , and a stage-two auxiliary entrance has rank $h^2 - h$. Thus the three classes listed above exhaust the ranks exceeding $m/2$, although exhaustion is not needed for the certificate: leaving any other edge unbatched can only weaken the claimed upper bound.

Apply the large-projector batching lemma to these classes. A rank- a_i projector with $a_i > m/2$ uses $m - a_i$ singleton pivot interchanges and one contiguous interchange of $(2a_i - m)$ subfields. The resulting table is

class	copies B_i	rank a_i	singleton pivots	batched width k_i
stage 1–3 join	$v^2 R$	21896	56	21840
stage 2 sink	$v^2 R$	21168	784	20384
stage 3 data entrance	$2v^3$	21141	811	20330

Treat every rank unit outside these three middle blocks as a singleton interchange. The number of singleton calls is therefore exactly

$$S = s - \sum_{i=1}^3 B_i k_i.$$

At a parent width e , the child widths are e/m with multiplicity S and $k_i e/m$ with multiplicity B_i ; all children operate on streams of volume V/W . The finite mixed-width recurrence consequently has normalized moment

$$\Psi(\tau) = \frac{S}{W m^\tau} + \sum_{i=1}^3 \frac{B_i}{W} \left(\frac{k_i}{m} \right)^\tau.$$

At $\tau = 1$ this is $s/(Wm) = 1 - \eta$, where $\eta = 39/520019360$.

The intermediate middle-block certificate. This paragraph records the first batching step. The controlled basis in the following subsection increases its saving from $177/10^9$ to $246/10^9$, which is the value used in the final assembly. Set

$$a_b = \frac{177}{10^9}, \quad \tau = 1 - a_b.$$

Let $r_0 = 1/m$, $r_i = k_i/m$, and

$$w_0 = \frac{S}{Wm}, \quad w_i = \frac{B_i k_i}{Wm}.$$

Then $\sum_i w_i = 1 - \eta$ and $\Psi(1 - a_b) = \sum_i w_i \exp(a_b \log(1/r_i))$. Use the rational logarithm upper bounds

$$\log m < \frac{9997}{1000}, \quad \log \frac{196}{195} < \frac{512}{100000}, \quad \log \frac{14}{13} < \frac{7411}{100000}, \quad \log \frac{21952}{20330} < \frac{768}{10000}.$$

They follow from the positive series

$$\log x = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}, \quad z = \frac{x-1}{x+1},$$

using sixteen terms and remainder at most $2z^{33}/(33(1-z^2))$; for m , first extract its power of two. For $0 \leq t < 1$, $e^t \leq (1-t)^{-1}$. Calling the four displayed upper bounds ℓ_i , exact rational arithmetic gives

$$\Psi(1 - a_b) < \sum_{i=0}^3 \frac{w_i}{1 - a_b \ell_i} < 1.$$

The strict gap in this last bound is greater than 1.78×10^{-10} . All integers, rational logarithm enclosures, weights and the exact gap are supplied by `scripts/batched_bit_rank_moment.py` and its JSON certificate. Under the accompanying arbitrary-width recurrence lemma, this proves the interchange saving $a_b = 177/10^9$ for the unchanged finite scalar producer. A smaller role upper bound can be inserted directly into the same three-class formula.

3.2 A simultaneous basis with one additional contiguous pivot block

The three-class rank bound can be strengthened while keeping the finite scalar producer unchanged. The additional structure is used only for the stage-two auxiliary sink edges. The same rational change of coordinates preserves the needed corner-rank conditions on the other two large classes.

Put $F = \mathbb{Q}^h$, $V = \mathbb{Q}^H$, $H = h^2$, and $m = hH$. Reorder the tensor factors so that the original third factor is F and the first two factors form V . List coordinates in the order

$$(\alpha, i), \quad 1 \leq \alpha \leq h, \quad 1 \leq i \leq H,$$

with i varying fastest. Write Π_t and Π_q for rank-one idempotents, and Π_U for a rank- h idempotent. The three large projector classes then have the following descriptions:

$$\begin{aligned} A_1 &= I - I_F \otimes \Pi_q - \Pi_t \otimes \Pi_U, & \Pi_q \Pi_U &= \Pi_U \Pi_q = 0, \\ A_2 &= (I_F - \Pi_t) \otimes I_V, \\ A_3 &= (I_F - \Pi_t) \otimes (I_V - \Pi_q). \end{aligned}$$

Their nullities are $2h$, H , and $H + h - 1$, respectively. For the first class, q spans the line associated with $t_B \otimes t_{\pi(A)}$, $U = F \otimes \langle t_A \rangle$, and $t = t_B$. The orthogonality in the bank matching gives the displayed zero products and, in particular, $q \notin U$. The second class is the stage-two sink projection. The third is the stage-three data entrance projection. Transposing any of these matrices preserves its displayed form, with the corresponding dual rank-one lines and subspaces.

Choose $K \in \text{GL}(V)$ and independent matrices $G_i \in \text{GL}(F)$ for $1 \leq i \leq H$. Define

$$T(u \otimes e_i) = G_i u \otimes e_i, \quad S = T(I_F \otimes K).$$

Thus T changes the h first-factor coordinates independently over each of the H multiplicity coordinates. The parameters are rational matrices; S is a change of basis in the finite address-label space, not a tape operation that must be performed on the input streams.

Lemma 3.1 (Simultaneous controlled basis). *For the finite collection of projectors in the retained $h = 28$ motif, one can choose $K, G_1, \dots, G_H$ over $\mathbb{Q}$ so that:*

1. *for every A_1 , the submatrix of SA_1S^{-1} in its first $2h$ rows and last $2h$ columns is nonsingular;*
2. *for every A_3 , the submatrix of SA_3S^{-1} in its first $H + h - 1$ rows and last $H + h - 1$ columns is nonsingular;*
3. *for every A_2 , the submatrix of SA_2S^{-1} in its first H rows and last H columns is diagonal and nonsingular.*

Proof. First consider the third assertion. The factor $I_F \otimes K$ commutes with A_2 . At multiplicity coordinate i , the transformed operator is $G_i(I_F - \Pi_t)G_i^{-1}$. Consequently its first-to-last block is diagonal in i , with diagonal entry

$$[G_i(I_F - \Pi_t)G_i^{-1}]_{1,h}.$$

For a nonzero rank-one projector and $h \geq 2$, this entry is a nonzero rational function of G_i : writing $\Pi_t = t\varphi$ with $\varphi(t) = 1$, it equals $-(e_1^\top G_i t)(\varphi G_i^{-1} e_h)$, and the two factors can both be made nonzero. Thus the required diagonal entries can all be required to be nonzero.

For the first assertion, let $Q_1 = I - A_1$. Its image is the direct sum

$$\text{im } Q_1 = (F \otimes q) \oplus (t \otimes U).$$

Choose a basis matrix B of the h -space U . After applying K , write $q' = Kq$ and $B' = KB$. The nullspace vector parametrized by $u \in F$ and $c \in \mathbb{Q}^h$ has multiplicity- i coordinate

$$G_i(q'_i u + tB'_i c).$$

The first $2h$ coordinate rows therefore evaluate

$$q'_i \ell_i(u) + \ell_i(t)B'_i c, \quad \ell_i = e_1^\top G_i, \quad 1 \leq i \leq 2h.$$

We show that their determinant is not identically zero in the parameters. Choose K so that $q'_i \neq 0$ for $1 \leq i < h$ and the submatrix $[q' \ B']$ in rows $h, \dots, 2h$ is nonsingular. Such a K exists because $[q \ B]$ has column rank $h + 1$ and an invertible linear map can send this ordered independent system to any other one. On the first $h - 1$ rows, choose ℓ_i to be a basis of the annihilator of t . These rows determine u modulo $\langle t \rangle$. On the remaining $h + 1$ rows, choose one fixed functional ℓ with $\ell(t) = 1$. After the first $h - 1$ coordinates of u are determined, these rows form exactly the nonsingular matrix $[q' \ B']$ on the remaining coordinate of u and c . Every prescribed nonzero ℓ_i extends to the first row

of an invertible G_i . This proves that restriction of $\text{Sim } Q_1$ to the first $2h$ coordinates can have full rank.

For the second assertion, the image of $Q_3 = I - A_3$ is

$$\text{im } Q_3 = (\langle t \rangle \otimes V) \oplus (W_t \otimes \langle q \rangle), \quad W_t = \text{im}(I_F - \Pi_t), \quad \dim W_t = h - 1.$$

Since the first summand contains $t \otimes V$, its image under $I \otimes K$ is unchanged. Write a vector after that change as $tv_i + q'_i u$ at multiplicity coordinate i , where $v \in \mathbb{Q}^H$ and $u \in W_t$. The first H coordinate rows give

$$\ell_i(t)v_i + q'_i \ell_i(u), \quad 1 \leq i \leq H.$$

Choose all $\ell_i(t) \neq 0$. They determine the H independent coordinates v_i once u is known. The next $h - 1$ rows are the second rows of G_i at $i = 1, \dots, h - 1$. If $f_i = e_2^\top G_i$, eliminating v_i gives

$$q'_i \left(f_i - \frac{f_i(t)}{\ell_i(t)} \ell_i \right) (u).$$

Choose K with these q'_i nonzero and choose the displayed $h - 1$ functionals to be a basis of the annihilator of t , restricted to W_t . This is possible because $F = \langle t \rangle \oplus W_t$: take each $\ell_i(t) = 1$, take the relevant f_i to annihilate t , and complete these independent first two rows to G_i . Thus restriction to the first $H + h - 1$ coordinates can also have full rank.

Apply the same two arguments to the transposed projectors at the last coordinate rows. This is a condition within the same parameter family, because

$$S^{-\top} = T^{-\top}(I_F \otimes K^{-\top}),$$

and $T^{-\top}$ acts independently by $G_i^{-\top}$. They show that the corresponding last-coordinate restrictions of the transformed dual nullspaces can each have full rank.

To combine the requirements, express every rank condition as a nonzero minor after clearing the denominators of the parameter matrices and their inverses. Each is a polynomial that is not identically zero, by the constructions just given. Their finite product, together with the invertibility conditions on K and all G_i , is still a nonzero polynomial. A nonzero rational polynomial cannot vanish at every rational tuple, so one rational choice meets all conditions simultaneously. The individual witness choices above establish that the polynomials are nonzero; they need not themselves meet all the other conditions.

Finally, for a rank- r nullspace projector $Q = UV$ with $VU = I_r$, its first- r by last- r submatrix after conjugation factors as the product of the two square restriction matrices just considered. Both are nonsingular. The corresponding block of $I - SQS^{-1}$ is its negative, since these first and last index sets are disjoint. This proves the first two assertions. The third has already been proved directly. $\square$

The argument is constructive in the finite-data sense required by the network interface: enumerate rational parameter matrices and retain the first choice passing the finitely many specified determinant tests. The lemma proves termination. All parameters and tests depend only on the fixed motif, not on the eventual multiplication input size. Conjugate every terminal and gate matrix by this one S . Their endpoint differences remain I_m , common gate matrices remain common, and all edge ranks are preserved.

The improved stage-two pivot profile. The lower-triangular elimination for a large projector first eliminates its nonsingular corner block. For A_2 , this block is already diagonal and its H pivots lie in matching physical order in two contiguous sets of fields. After fixed diagonal normalizations, they are one $H \times H$ identity block. The Schur-complement argument in the large-projector lemma leaves the middle identity block of size $m - 2H$. Hence a stage-two sink edge is implemented by two contiguous child interchanges, of widths

$$Hb \quad \text{and} \quad (m - 2H)b,$$

with no singleton recursive calls on its corner. Its rank remains $H + (m - 2H) = m - H$, so the rank sum at exponent one is unchanged. The first and third edge classes retain their previous singleton-corner and contiguous-middle implementations by the first two parts of the lemma.

Exact strengthened moment. Let $B = v^2 R$ and let S_0 be the singleton count from the three-class certificate. Replace S_0 by $S_0 - BH$ and add B children of width H/m . For $h = 28$ the new singleton count is

$$S_1 = 105555640096680883200.$$

The normalized child widths and their rank-mass weights are respectively

$$\left(\frac{1}{21952}, \frac{195}{196}, \frac{13}{14}, \frac{10165}{10976}, \frac{1}{28} \right)$$

and

$$\left(\frac{10841531}{520019360}, \frac{12827685}{26000968}, \frac{855179}{1857212}, \frac{20865}{2736944}, \frac{65783}{3714424} \right).$$

Use the four preceding rational logarithm bounds and additionally $\log 28 < 33323/10000$. With

$$a_b = \frac{246}{10^9}, \quad \tau = 1 - a_b,$$

the same bound $e^x \leq (1 - x)^{-1}$ gives a strict rank-moment gap exceeding 4.76×10^{-11} . The exact rational calculation is in `scripts/controlled_bit_rank_moment.py`. This proves the stronger interchange saving using the unchanged PR #7 scalar producer and the same arbitrary-width recurrence interface.

3.3 Integer widths, bounded row padding, and mixed contractions

The batched residual construction uses a fixed finite list of child sizes, not only powers of one radix. The following implementation supplies the corresponding integer-width recursion. All constants in this subsection are fixed finite circuit constants, independent of the input length.

A general moment condition. Let $m, W \geq 2$ be fixed. At one circuit invocation, let the list $t_1, \dots, t_J$ contain integers in $[1, m - 1]$. A child of type i has width $t_i f$, where $f = \lfloor e/m \rfloor$, and operates on exactly $1/W$ of the parent's logical volume. Repetitions in the list represent separate calls. An ordinary rank-one call has $t_i = 1$; a batched block has the corresponding larger value of t_i . Full-rank edges may retain their original m rank-one calls, so no same-width recursive call is needed. Put

$$\Psi(x) = \frac{1}{W} \sum_{i=1}^J (t_i/m)^x, \quad t_* = \max_i t_i < m.$$

If $\Psi(\tau) < 1$ for some $0 < \tau < 1$, then the normalized recurrence

$$F(e) \leq \frac{1}{W} \sum_{i=1}^J F(t_i \lfloor e/m \rfloor) + C, \quad F(e) = O(1) \quad (1 \leq e < m),$$

has $F(e) = O(e^\tau)$ for every positive integer e . Indeed, substituting $F(u) \leq Au^\tau$ into an internal call gives at most $A\Psi(\tau)e^\tau + C$. Choose A to cover the finite base range and $C/(1 - \Psi(\tau))$. Floors can only decrease the right-hand side.

A finite integer depth bound. Define

$$D(e) = 0 \quad (e < m), \quad D(e) = 1 + D(t_* \lfloor e/m \rfloor) \quad (e \geq m).$$

This is an integer algorithm: repeatedly replace e by $t_* \lfloor e/m \rfloor$ until it is below m . The function D is nondecreasing, every child has depth bound at most $D(e) - 1$, and $D(e) = O(\log(2e))$, because each internal replacement is at most $(t_*/m)e$. A row count divisible by $W^{D(e)}$ is therefore sufficient for every root-to-leaf path. At each invocation write the row index as $Wg + w$ and split by w . Every child has complete within-row ranges, logical volume exactly V/W , and a remaining row count divisible by $W^{D(e)-1}$. No padding occurs inside this recursion.

Bit interchange with a remainder. At an integer-width call write $e = mf + r$, $0 \leq r < m$. If $f = 0$, interchange the bounded number of digits individually. Otherwise separate the high r digits of each of the two target chunks. With G denoting all intervening spectators, the layout is

$$(H^+, H^-, G, D^+, D^-).$$

First exchange corresponding digits of H^+, D^+ and move H^+ just before H^- . The layout becomes (D^+, H^+, H^-, G, D^-) . Apply the finite circuit to the two length- mf chunks H^-, D^- . Finally move H^+ back immediately before the final H^- , obtaining

$$(D^+, D^-, G, H^+, H^-).$$

All other fields are preserved. These moves use fewer than a fixed multiple of m single-digit operations and cost $O(V)$. The main chunks split into m equal width- f fields, so every rational shear and every batched contiguous-block shear has exactly the required field shape. A child retains the unused main digits and all remainders as spectators. Thus the preceding moment recurrence applies literally at every integer width; there is no per-level numerical padding of either chunk range.

Supplying the bit rows from existing digits. For an arbitrary width e , compute $D(e)$ and choose the least nonnegative integer ρ such that

$$q^{2\rho} \geq W^{D(e)}.$$

Both numbers can be constructed by fixed-radix integer arithmetic. Since $D(e) = O(\log e)$, we have $\rho = O(\log e)$. For sufficiently large e , $2\rho < e$. The exceptional set of smaller widths is fixed and finite; individual-digit interchange there is absorbed by the asymptotic constant.

Exchange the high ρ digits of the two chunks, then gather their two high parts into one row field before both remaining chunks, exactly as in the preceding layout operation. This row field has range $q^{2\rho}$. Within each original prefix, pad its range to

$$R' = W^{D(e)} \left\lceil q^{2\rho} / W^{D(e)} \right\rceil < 2q^{2\rho}.$$

Every added row includes the entire remaining address set. Recurse once at the remaining width $e - \rho$, which has depth bound at most $D(e)$; then remove the added rows and undo the gathering move. Each completed invocation preserves its row index, so padded zero rows are zero again when they are removed.

The possibly large exponent in $W^{D(e)} = e^{O(1)}$ does not multiply the logical volume by that polynomial. The row digits were already part of the two input chunks. Only rounding the row count adds data, by a factor strictly below two, once at the outer boundary. Exchanging and gathering $O(\log e)$ high digits costs $O(V \log(2e))$, which is $O(Ve^\tau)$. The radix conversion from binary chunk ranges to the fixed radix q retains the original constant factor, less than q^2 . Consequently the batched construction is an arbitrary-width interchange procedure with the asserted exponent τ and no additional power of its volume or width.

Fixed tapes and descriptors. Use the original depth-first scheduler: park only the other role streams at the active stack top, run the child on one complete role stream, and restore the parked streams on return. The number of child types is fixed, so parking, copying, cleanup and the finite arithmetic transforms contribute $O(V)$ per node. Before descending, retain the two current chunks and merge consecutive spectator fields; the descriptor has a fixed number of field intervals. Its integer lengths have $O(\log V_0)$ bits. All descendants retain the complete ranges of both root main chunks, even when most digits have become spectators. Their volume is therefore at least $q^{2(e-\rho)}$ in the outer construction, while $j \log W = O(\log e)$ at depth j . Hence $\log V_0 = \log V_j + j \log W = O(\log(2V_j))$. Polynomial descriptor work is absorbed by $O(V_j)$, as in the original fixed-tape proof. The larger fixed constant in the depth bound creates no new tape and no growing alphabet.

4 Complex residuals and arithmetic depth

4.1 The weighted arithmetic-path budget

The precision analysis uses the length of an arithmetic dependency path through the framed complex network. We first measure this path in directional rank units, before grouping some units into bulk children.

A weighted path bound. Use the retained triple-based complex motif on h points, with $m = h^3$. This includes the paired complex producer and its shared first- and third-stage auxiliary banks. Regard every scalar gate as a map from saved inputs to its outputs. All of its inputs and outputs have one common subspace label. A directed dependency path can therefore enter a gate on one physical wire and leave on another without changing label dimension at the gate itself.

For a network edge a , let U_a^- and U_a^+ be its labels and write $r_a = |\dim U_a^+ - \dim U_a^-|$. The edge implementation has exactly r_a recursive calls, each implementing one directional C tensor on e/m selected axes. Binary changes of coordinates before and after these calls permute coefficients and add no arithmetic depth. An inverse direction is implemented using one forward child and its sign and fourth-root wrappers, so it has the same recursive-call count.

If P is a directed scalar dependency path, its common gate labels give the exact identity

$$\sum_{a \in P} r_a = \dim U_{\text{end}} - \dim U_{\text{start}} + 2 \sum_{\substack{a \in P \\ \dim U_a^+ < \dim U_a^-}} (\dim U_a^- - \dim U_a^+).$$

The endpoint difference is at most m .

Every side-producer edge is increasing. This follows from nested source labels for the nonconstant middle mixer, complemented labels in its reverse-direction counterpart, and constant early and late frames. The data-wire and interstage edges are increasing as in the retained motif. The stage-sharing joins are also increasing. The only decreasing edges are the retained central returns, each of dimension loss h .

There are three tensor stages. Within a stage, the v^2 invocations use disjoint data and auxiliary wires, and each invocation has a single central-return frontier. Equivalently, its point and total central wires all return between the same two blocks of the transparent schedule. A directed path crosses that frontier at most once. Gates acting on different invocations do not connect their wires. Thus a path encounters at most one central return per stage, even though the machine executes all invocations sequentially on its fixed tapes. Sharing stage-1 and stage-3 banks only links their existing stage boundaries and creates no additional return. Consequently

$$\sum_{a \in P} r_a \leq q := m + 6h. \quad (4.1)$$

The bound also applies to a path ending at an intermediate gate or inside an edge implementation: complete its current edge, and then any suffix to a terminal vertex. All weights are nonnegative.

From a scalar path to a coefficient path. A coefficient dependency path in the actual tape algorithm projects to a directed path of the scalar network: inside a scalar gate it follows one of the saved operands that contributes to the chosen output; inside an edge it stays on the edge's role stream. A child can permute or combine addresses within that role, but cannot move a value into a different parent role. The row split, parking, padding and reassembly operations do not create arithmetic dependencies between distinct parent roles. This projection justifies using (4.1) for the implemented recursion, rather than merely for a drawn finite circuit.

4.2 A whole-residual complex child

The complex phase interface admits a further batching step. On an edge with residual dimension a , the existing binary change of basis puts an orthonormal residual basis into the first a slots. If each slot has f selected axes, those first a slots form one contiguous interval of af axes. Instead of a child calls on f axes, use one child on that interval.

The directions need not all have the same sign. A basis vector's weight modulo four may make its directional kernel an inverse even on an increasing edge. Let B be the set of negative residual slots and let Z_B multiply an address by -1 precisely when the parity of its selected bits in those slots is odd. The exact identity $C^{-1} = -iZCZ$ gives

$$\bigotimes_{j=1}^a (C^{\varepsilon_j})^{\otimes f} = (-i)^{f|B|} Z_B C^{\otimes af} Z_B, \quad \varepsilon_j \in \{1, -1\}.$$

Thus one forward child, two address-sign passes and one fixed fourth-root phase implement the mixed signs. The same binary basis changes before and after the child are retained. The wrappers

cost $O(V)$: their sign is computed from the current address descriptor, whose polynomial work per record is absorbed by the existing superpolynomial record length. They introduce at most four elementary coefficient operations per value and no further recursive call.

We batch only the two specified auxiliary edge classes of ranks

$$a_1 = m - 2h, \quad a_2 = m - h^2.$$

All other edges continue to use their old one-direction children. The first class is the increasing join between a completed stage-1 auxiliary bank, at dimension h , and its matched stage-3 bank, at dimension $m - h$. The second is the stage-2 auxiliary exit from dimension h^2 to the full terminal frame. These are complete auxiliary banks, including the central roles when those are shared. If a width bound requires padding, add dummy auxiliary roles with identity scalar gates at the ordinary low and high frames before matching the first- and third-stage banks. Their labels only increase, their arbitrary scalar values are restored, and both counted auxiliary classes are then present for every padded role. The changed volume-weighted time expression is consequently

$$\frac{1}{W} \left[(s - n_1 a_1 - n_2 a_2) m^{-\sigma} + n_1 \left(\frac{a_1}{m} \right)^\sigma + n_2 \left(\frac{a_2}{m} \right)^\sigma \right],$$

where n_1, n_2 count the chosen edges. It has value $s/(Wm) < 1$ at $\sigma = 1$. Strict comparison below one supplies the new complex recursion exponent. The separate mixed-width row construction handles $f = \lfloor e/m \rfloor$, the at most $m - 1$ individually processed remaining axes, and a row reservation sufficient for the longer but still $O(\log d)$ recursion.

4.3 A depth guard for the unequal complex children

The dependency-path argument of Section 4.1 still bounds the sum of edge ranks on one scalar path by

$$q = m + 6h.$$

For $h = 28$ its relevant constants are

$$m = 21952, \quad q = 22120, \quad a_1 = 21896, \quad a_2 = 21168.$$

Both selected ranks exceed $q/2$. Therefore every arithmetic dependency path encounters at most one selected bulk edge. At a call on e axes, put $f = \lfloor e/m \rfloor$. Ignoring the local additive operations, the path has one of the following child-depth bounds:

$$qA(f), \quad (q - a_1)A(f) + A(a_1 f), \quad (q - a_2)A(f) + A(a_2 f).$$

The weight budget accounts for both increasing and decreasing frame edges and for paths that switch wires at a scalar gate. Inverse-child and endpoint sign corrections add no child. Individually processing the fewer than m leftover axes adds at most $8m$ elementary operations, absorbed into the local constant below.

Take $\rho = 6/5$. The following three normalized path expressions have a rational common upper bound strictly below one:

$$\max \left\{ \frac{q}{m^\rho}, \quad \frac{q - a_1}{m^\rho} + \left(\frac{a_1}{m} \right)^\rho, \quad \frac{q - a_2}{m^\rho} + \left(\frac{a_2}{m} \right)^\rho \right\} < \bar{\theta} := \frac{999}{1000}. \quad (4.2)$$

Here is an entirely rational verification. First $160000^5 < m^6$, so $m^\rho > 160000$. For $0 < t < 1$, Taylor's formula with its second derivative bounded on $[0, t]$ gives

$$(1-t)^{6/5} \leq 1 - \frac{6}{5}t + \frac{3}{25} \frac{t^2}{1-t}.$$

Indeed its second derivative is $(6/25)(1-t)^{-4/5} \leq (6/25)/(1-t)$. Putting $t = (m - a_i)/m$ gives upper bounds, in the displayed order,

$$\frac{553}{4000}, \quad \frac{47817969}{47897500}, \quad \frac{1213697}{1260000},$$

all strictly below 999/1000.

Stopping and exact arithmetic. Retain a global stopping threshold d^β , with $0 < \beta < 1$. For sufficiently large d , this threshold exceeds $2m$. A leaf reached from an internal call has width at least $d^\beta/(2m)$: every child has width at least $f = \lfloor e/m \rfloor \geq e/(2m)$. Put

$$E = 64(W + m + 1)^3, \quad C_{\text{dep}} = 1000(E + 16m + 1).$$

The exact finite inequality $36W^3 + 4s + 4W + 8m + 4 < E$ covers all local scalar operations, the new leftover-axis work and the retained endpoint and inverse wrappers. For every descendant width e reached before or just after stopping, strong induction gives

$$A(e) \leq C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho. \quad (4.3)$$

At a leaf, use

$$8e \leq 8(2m)^{\rho-1} d^{-\beta(\rho-1)} e^\rho \leq 16m d^{-\beta(\rho-1)} e^\rho.$$

At an internal call, child widths are at most their stated fractions of e , so (4.2) bounds their total by $\bar{\theta} C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho$. Since $e \geq d^\beta$, the remaining gap is at least $(1 - \bar{\theta}) C_{\text{dep}} d^\beta \geq E$. This proves the induction. A root that is already a leaf separately satisfies $A(e) \leq 8d^\beta$, which is smaller than the final bound below.

For rational $\zeta > 0$, set

$$C_1 = \rho - (\rho - 1)\beta + \zeta, \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil, \quad \Delta = \lceil C_0 d^{C_1} \rceil.$$

The root bound from (4.3) is $C_{\text{dep}} d^{\rho-(\rho-1)\beta}$. This covers the direct arbitrary-width root construction; it also covers every retained base- m piece. Even retaining the older bound of $m(1 + 1/\zeta)d^\zeta$ such pieces, their concatenation, all reserved axis kernels, the outer phases and the b^D conversion have depth at most $C_0 d^{C_1}$. The original elementary-depth induction therefore supplies exact denominator and magnitude guards of $p + \Delta$ fractional bits and $\Delta + 3$ integer/sign bits. The width remains $O(p)$ when $\epsilon C_1 < 1$.

In particular, $\beta = 1/1000$ and $\zeta = 1/10000$ give

$$C_1 = \frac{11999}{10000}, \quad 1 - \frac{7999999}{16000000} C_1 = \frac{64008011999}{160000000000} > 0.$$

Thus bulk complex children preserve the near-half dimension exponent while allowing the improved complex time exponent to be used with almost no loss at the stopping leaves.

5 The complete layer on mixed widths

5.1 The complex layer with unequal recursive axis counts

Apply the same integer-depth construction to the complex circuit's list of child sizes. For a selected residual of rank $a < m$, the residual orthonormal basis places its nontrivial coordinate slots first. Its a consecutive slots of f selected axes form one contiguous block of af axes. The residual factors can have signs $\varepsilon_v \in \{1, -1\}$. For their negative slot set B , the retained identity $C^{-1} = -iZCZ$ gives, after the residual basis map,

$$\prod_{v=1}^a (C^{\varepsilon_v})^{\otimes f} = (-i)^{|B|} Z_B C^{\otimes af} Z_B.$$

Here Z_B applies Z on the f axes of each negative slot. Thus one recursive application of $C^{\otimes af}$, two address-sign passes, and one exact phase replace the a separate recursive calls; the basis maps before and after the operation are unchanged. The sign descriptors have polynomial size in the global parameters and fit the existing per-record overhead allowance. Other residuals retain their original rank-one calls. Let Ψ_c and D_c denote the resulting moment and depth functions. In particular, the two bulk classes of the current motif have

$$a_1 = m - 2h = 21896, \quad a_2 = m - h^2 = 21168 \quad (m = 21952, h = 28),$$

so every recursive call is a strict contraction.

Complete rows and compact fields. Keep the original global p, d, K, G, H, q_F, q_B for compact control, and replace only the row reservation by

$$k_0 = D_c(d), \quad q_0 = \lceil \log_2 W \rceil k_0, \quad Q_0 = q_0 + q_F + q_B.$$

This conservative integer choice satisfies $2^{q_0 K} \geq W^{k_0}$ and $q_0 = O(\log(2d))$. If the available number D of selected axes is at most Q_0 , apply them individually. Otherwise process the first $q_0 + q_F$ and last q_B axes individually and use the remaining consecutive middle block for the recursive circuit. The row field consists only of the first q_0 chunks; neither compact field is included in it. Pad its count to the next multiple of W^{k_0} , by a factor below two, once before recursion. Every child therefore has all compact and spectator ranges and exactly $1/W$ of its parent's volume, with enough row divisibility for its remaining depth.

The reserved-axis bound is unchanged:

$$Q_0 = O(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1).$$

In particular, the larger fixed row-depth constant does not change any power of d or p in preprocessing. The exact bit interchange called inside a compact operation supplies its own rows from its own target chunk digits; its internal row bound is not an additional reservation of selected complex axes.

Integer axis counts. At a complex call of width $e = mf + r$, first apply the $r < m$ leftmost selected kernels individually and treat those axes as completed spectators. The remaining mf axes are consecutive and split into the required m equal slots. The extra work is $O(V)$ per node. The individual kernels on removed axes commute with the completed tensor map on the other axes, so this produces the desired map on all e axes. The root active block can be processed directly; its base- m expansion is unnecessary.

Stopping and weighted-tree bounds. Fix rational $0 < \beta < 1$ and stop at $e < d^\beta$, with normalized leaf cost $O(e)$. Suppose $\Psi_c(\sigma) < 1$ for $0 < \sigma < 1$. Each node's normalized volume weight is W^{-j} at depth j . Its children have total σ -weighted width at most $\Psi_c(\sigma)$ times the parent's weighted width. Therefore

$$\sum_{\text{leaves } u} W^{-\text{depth}(u)} e_u^\sigma \leq e_{\text{root}}^\sigma.$$

Since $e_u < d^\beta$, total normalized leaf work is at most

$$O(d^{\beta(1-\sigma)} e_{\text{root}}^\sigma) = O(d^{\sigma+\beta(1-\sigma)}).$$

This is a bound for a tree with unequal child widths; no common leaf depth is assumed.

Compact operations cost $O(G^\tau e^\tau + 1)$ per current logical volume. If $\sigma \leq \tau$, monotonicity gives $\Psi_c(\tau) \leq \Psi_c(\sigma) < 1$, and summing the τ -weighted widths by depth bounds all internal work by $O(G^\tau d^\tau)$. If $\sigma > \tau$, every internal width is at least d^β , so $e^\tau \leq d^{\beta(\tau-\sigma)} e^\sigma$; summing the strict σ -moment gives $O(G^\tau d^{\sigma+\beta(\tau-\sigma)})$. Thus in both cases, with

$$\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\},$$

the complete normalized layer cost is

$$O((\log p)^3 [d^\chi + d^{\sigma+\beta(1-\sigma)} + d^{\max\{1-c, 0\}} + 1]).$$

The displayed logarithmic factor conservatively retains the existing layout and descriptor allowances. Hence the original strict conditions

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda'$$

continue to imply the complete $O(Vd^{\lambda'})$ layer bound.

Compatibility with the new dependency guard. The separate dependency-path estimate uses $q = m + 6h = 22120$, observes that one path crosses at most one selected bulk residual, and bounds its ρ -moment by

$$\max \left\{ \frac{q}{m^\rho}, \quad \frac{q - a_1}{m^\rho} + \left(\frac{a_1}{m} \right)^\rho, \quad \frac{q - a_2}{m^\rho} + \left(\frac{a_2}{m} \right)^\rho \right\} < \frac{999}{1000} \quad \left(\rho = \frac{6}{5} \right).$$

Integer floors only shrink these children. For sufficiently large d , an internal parent has $e \geq d^\beta \geq 2m$, hence every recursive child has width at least $\lfloor e/m \rfloor \geq d^\beta/(2m)$ whenever it becomes a stopping leaf. The at most $m - 1$ individually processed remainder axes at each internal node belong to its fixed local overhead; add their bounded coefficient contribution to the guard's fixed constant E . They are not extra arbitrarily small stopping leaves. Thus the separately established guard exponent

$$C_1 = \rho - (\rho - 1)\beta + \zeta = \frac{6}{5} - \frac{\beta}{5} + \zeta$$

applies to this integer-width implementation. All stopping tests can be performed by the existing rational-power integer comparison.

6 The batched baseline assembly of PR #10

6.1 An exact complex exponent after bulk recursion

Keep the paired complex producer of PR #7. Its constants are

$$h = 28, \quad v_c = \binom{28}{3} = 3276, \quad m = 21952, \quad R_c = 93838.$$

There are $R_c + h + 1 = 93867$ auxiliary roles per invocation, including the original central roles. Write

$$\begin{aligned} B_c &= v_c^2(R_c + h + 1) = 1007397164592, \\ W_c &= 2v_c^2(v_c + R_c + h + 1) = 2085111546336, \\ L_c &= 3v_c^2h(h + 1) = 26143580736, \quad s_c = W_cm - 2v_c^3 + 2L_c = 45772350635112192. \end{aligned}$$

The two selected whole-residual classes have ranks $a_1 = 21896$ and $a_2 = 21168$, each with B_c copies. The number of remaining singleton children is

$$S_c = s_c - B_c(a_1 + a_2) = 2389799139122304.$$

Their time moment is

$$\Xi(\sigma) = \frac{S_c}{W_cm^\sigma} + \frac{B_c}{W_c} \left[\left(\frac{a_1}{m} \right)^\sigma + \left(\frac{a_2}{m} \right)^\sigma \right].$$

At $\sigma = 1$, this equals $1 - \eta_c$ with $\eta_c = 5/12693352$.

Take

$$a_c = \frac{7}{10^7}, \quad \sigma = 1 - a_c.$$

The same positive logarithm series used in the bit certificate gives

$$\log m < \frac{9997}{1000}, \quad \log \frac{m}{a_1} = \log \frac{392}{391} < \frac{2555}{10^6}, \quad \log \frac{m}{a_2} = \log \frac{28}{27} < \frac{36368}{10^6}.$$

Let these upper bounds be ℓ_0, ℓ_1, ℓ_2 and put

$$w_0 = \frac{S_c}{W_cm}, \quad w_i = \frac{B_c a_i}{W_cm} \quad (i = 1, 2).$$

The exact rational comparison is

$$\Xi(1 - a_c) < \sum_{i=0}^2 \frac{w_i}{1 - a_c \ell_i} < 1.$$

The gap in the last inequality is

$$\frac{24132873235669673777179671369131}{1525632551364197359955324563293421369131} > 1.58 \times 10^{-8}.$$

This certifies the stronger complex exponent with the unchanged finite producer and the new whole-residual recursion.

6.2 Final multiplication parameters and margins

Use the controlled-basis projector-batched interchange saving $a_b = 246/10^9$ and the preceding complex saving $a_c = 7/10^7$. The complete rational parameter choice is

$$\begin{aligned} \tau &= 1 - \frac{246}{10^9}, & \sigma &= 1 - \frac{7}{10^7}, & \epsilon &= \frac{7999999}{16000000}, \\ c &= 1, & \beta &= \frac{1}{1000}, & \delta &= \frac{1}{10^{10}}, \\ \zeta &= \frac{1}{10000}, & C_1 &= \frac{11999}{10000}, \\ \lambda &= 1 - \frac{246}{10^9} + \frac{1}{10^{16}}, & \lambda' &= 1 - \frac{246}{10^9} + \frac{2}{10^{16}}, \\ \kappa &= \frac{6149999}{5 \cdot 10^{13}}. \end{aligned}$$

The value $c = 1$ is allowed by the compact-control construction. Its reservation exponent $\max\{1 - c, 0\}$ is zero, while $1 - \epsilon(1 + c) = 1/8000000 > 0$ still ensures $K = o(\ell)$ for $K = d$. The larger fixed constants in the mixed-width row reservations do not alter any power of p .

Since $\sigma < \tau$, the compact-control internal exponent is τ . Its leaf exponent is

$$\sigma + \beta(1 - \sigma) = 1 - \frac{6993}{10^{10}} < \tau.$$

The strict gaps of 10^{-16} between τ, λ, λ' absorb the fixed powers of $\log p$ in the layer construction. The new coefficient guard satisfies

$$1 - \epsilon C_1 = \frac{64008011999}{160000000000} > 0.$$

The fast Gaussian construction retains $\alpha = \lfloor \sqrt{b/(8d)} \rfloor$ and $\gamma = 2d\alpha^2 \leq b/4$. Here $1 - 2\epsilon = 1/8000000 > 0$, so its eventual hypotheses $b \geq 64d^2$, $\alpha^2\theta_i \geq 1$ and the prime-interval growth condition all continue to hold. The numerical maps, their target errors, the final rounding step and the input/output format are unchanged. For the retained chirp input generator, use $F = 2^{\lceil 1.14\alpha^2 \rceil}$ to bound both magnitude and shifted absolute error; the corrected block argument still uses $P = 34p$.

The seven margins below one in the assembly cost table are

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), & g_3 &= \epsilon(1 - \lambda'), \\ g_4 &= (1 - \epsilon)(1 - \tau), & g_5 &= 1 - \delta - 2\epsilon, & g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon. \end{aligned}$$

Their exact values are

margin	value
g_1	$1/8000000$
g_2	$983999877/8000000000000000$
g_3	$9839998762000001/80000000000000000000$
g_4	$984000123/8000000000000000$
g_5	$1249/10000000000$
g_6	$312500039/625000000$
g_7	$7999999/16000000$

All twenty-nine parameter constraints are strict. The minimum final margin is $G = g_3$, and

$$G - \kappa = \frac{362000001}{800000000000000000000} > 0.$$

This last fixed positive gap absorbs the remaining logarithmic factors. The resulting conditional multiplication bound is

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{6149999}{5 \cdot 10^{13}} = 1.2299998 \times 10^{-7} > 2^{-23}.$$

The exact comparisons are reproduced by `scripts/batched_network.py` and recorded in `certificates/batched-network.json`.

7 Auxiliary source frames in the interchange network

7.1 Auxiliary source frames

The transfer construction assigns a rational matrix to every source, gate and sink, common to all incidences of a gate, and requires

$$M_{\text{sink}(\rho(w))} - M_{\text{source}(w)} = I_m$$

for every input role w . The retained networks give every auxiliary source the zero matrix and every auxiliary sink the identity. The transfer does not need this choice.

Lemma 7.1 (Auxiliary source frames). *Let w be an auxiliary role, so that $\rho(w) = w$, and let M be a fixed rational $m \times m$ matrix. Assign M to the source of w and $I_m + M$ to its sink, and leave every other matrix unchanged. Then the scalar network, every common gate matrix and every endpoint identity are preserved, and the address map implemented on every role is unchanged. Only the first and last edges of w change.*

Proof. The endpoint identity for w reads $(I_m + M) - M = I_m$; no other endpoint identity involves this source or sink. The physical input stream f_w has the logical interpretation $g_w = \Phi_{-M} f_w$. This is a definition, not a tape operation, and g_w is arbitrary because f_w is. The scalar network restores arbitrary auxiliary values, so the logical value at the sink is again g_w . The common-frame identity gives the physical output

$$\Phi_{I_m+M} \Phi_{-M} f_w = \Phi_{I_m} f_w,$$

the same shear as on every other role. Gates on w keep their common matrices, so every interior edge of w , and every edge of every other role, is unchanged. $\square$

The lemma moves rank between the two end edges of an auxiliary role. Before batching only the total mattered. With contiguous blocks, a large edge costs much less per rank unit than a small one, so concentrating a role's rank in one edge lowers the cost.

Stage-two auxiliary roles. Use the notation of Section 3.1. At tensor stage two put $A = F$, $P = \langle t_{a_1} \rangle$, $B = P^\perp$ and $Q = \langle t_{a_3} \rangle$. Every auxiliary role of a stage-two invocation, including its central roles, has first gate frame $D_0 = (B \otimes F) \otimes Q$ of dimension $h^2 - h$, and last gate frame $D_1 = (A \otimes F) \otimes Q$ of dimension h^2 . Assign every such role the source matrix P_{D_0} , and therefore the sink matrix $I + P_{D_0}$. Then:

1. the entrance edge has matrix $P_{D_0} - P_{D_0} = 0$ and makes no call;
2. every interior edge is unchanged;

3. since $D_0 \subset D_1$ are nondegenerate, $D_0 \perp D_1^\perp$, and the exit edge has matrix

$$I + P_{D_0} - P_{D_1} = P_{D_1^\perp} + P_{D_0} = P_{D_0 \perp D_1^\perp}.$$

This is an idempotent of rank $m - h$. Its nullspace is $D_1 \cap D_0^\perp = P \otimes F \otimes Q$, of dimension h , because $A = B \perp P$.

Formerly the entrance had rank $h^2 - h$ and the exit rank $m - h^2$. Now the entrance has rank zero and the exit rank $m - h$. The total rank of each role, the rank sum s and the deficit $Wm - s$ are therefore unchanged. The stage-two sink class A_2 of Section 3.2 no longer occurs. Its replacement is

$$A_4 = I - \Pi_{P \otimes F \otimes Q},$$

where Π_U denotes the orthogonal projector onto U for the label form.

Lemma 7.2 (A common basis for the new exit class). *In the parameter family $S = T(I_F \otimes K)$ of Lemma 3.1, there are rational $K, G_1, \dots, G_H$ satisfying its first two assertions and, in addition: for every A_4 , the submatrix of SA_4S^{-1} in its first h rows and last h columns is nonsingular.*

Proof. Order coordinates as in Section 3.2: the original third factor F is slow and V , the first two factors, is fast. Then the nullspace of A_4 is $\langle t \rangle \otimes X$ with $t = t_{a_3}$ and $X = \langle t_{a_1} \rangle \otimes F \subset V$, of dimension h . Write $\Pi = UV'$ with $V'U = I_h$. Its first- h -row by last- h -column block after conjugation is the product of the first h rows of SU and the last h columns of $V'S^{-1}$. The corresponding block of SA_4S^{-1} is its negative, because these row and column index sets are disjoint. It is therefore enough that both square factors can be nonsingular. Each is then a polynomial condition that is not identically zero.

A nullspace vector $t \otimes x$, with $x \in X$, has coordinate $(G_i t)_\alpha (Kx)_i$ at (α, i) after applying S . The first h coordinates have $\alpha = 1$ and $1 \leq i \leq h$, so the first factor is the map $x \mapsto ((G_i t)_1 (Kx)_i)_{i \leq h}$. Give every G_i the first row $(1, \dots, 1)$, so that $(G_i t)_1 = |a_3| = 5$, and choose K sending a basis of X to $e_1, \dots, e_h$. The map is then nonsingular.

The rows of V' span the form dual $\Omega(\langle t \rangle \otimes X) = \langle \Omega_F t \rangle \otimes \Omega_V X$, where Ω is the tensor label form. This subspace has the same shape. Since $S^{-\top} = T^{-\top}(I_F \otimes K^{-\top})$ and $T^{-\top}$ acts by $G_i^{-\top}$, the same argument at the last h coordinates, with $\alpha = h$ and $H - h < i \leq H$, shows that the second factor can be nonsingular.

As in Lemma 3.1, clear denominators and multiply these nonzero polynomials over the finitely many A_4 . Include the polynomials for its first two assertions and the invertibility of K and every G_i . A nonzero rational polynomial does not vanish at every rational point. Its third assertion concerned A_2 and is no longer needed. $\square$

By the large-projector factorization of Section 2.1, every exit edge A_4 is compiled as h singleton interchanges and one contiguous interchange of its middle $m - 2h$ fields. The first and third classes of Section 3.1 keep their singleton corners and contiguous middle blocks.

7.2 The interchange rank moment with source frames

Keep the retained counts

$$h = 28, \quad v = \binom{28}{5} = 98280, \quad m = 21952, \quad R = 11840940,$$

$$W = 2v^2(v + R), \quad s = Wm - v^3 + 2L,$$

and put $B = v^2R = 114371146876896000$, the number of stage-two auxiliary roles. The large classes are now

class	copies	rank	singleton pivots	batched width
stage 1–3 join	B	21896	56	21840
stage 3 data entrance	$2v^3$	21141	811	20330
stage 2 auxiliary exit	B	21924	28	21896

Every other rank unit is a singleton interchange. Relative to the controlled count S_1 of Section 3.2, each stage-two auxiliary role loses its entrance of $h^2 - h$ singletons and gains h exit corner pivots. Its former sink corner and middle blocks are replaced by the new exit. Hence

$$S' = S_1 - B(h^2 - 2h) = 22293445170300595200.$$

The normalized child widths are

$$r = \left(\frac{1}{21952}, \frac{195}{196}, \frac{10165}{10976}, \frac{391}{392} \right),$$

and their rank-mass weights are

$$w = \left(\frac{2289741}{520019360}, \frac{12827685}{26000968}, \frac{20865}{2736944}, \frac{25721153}{52001936} \right).$$

The weights sum to $1 - \eta$ with $\eta = 39/520019360$, so batching again preserves the rank deficit. The positive logarithm series used before give

$$\log m < \frac{9997}{1000}, \quad \log \frac{196}{195} < \frac{512}{10^5}, \quad \log \frac{10976}{10165} < \frac{768}{10^4}, \quad \log \frac{392}{391} < \frac{2555}{10^6}.$$

Put

$$a_b = \frac{154}{10^8}, \quad \tau = 1 - a_b.$$

With $e^x \leq (1 - x)^{-1}$ for $0 \leq x < 1$, exact rational arithmetic gives

$$\Psi(1 - a_b) < \sum_{i=0}^3 \frac{w_i}{1 - a_b \ell_i} < 1,$$

with gap

$$\frac{9704597579224556535778749633448639054699}{20663392005574270637817582110595278287793980851574} > 4.69 \times 10^{-10}.$$

The arbitrary-width recurrence of Section 3.3 therefore gives the interchange saving $a_b = 154/10^8$ for the unchanged PR #7 scalar producer.

8 A third complex residual class

8.1 A third whole-residual complex class

The whole-residual construction of Section 4.2 applies to any edge residual. We additionally select the stage-three data entrance edges. At stage three the incoming physical X and Y labels have

dimensions h^2 and $h^2 - 1$, and their first relevant frames have dimensions $(h^2 - 1)h + 1$ and $(h^2 - 1)h$. Both entrance residuals are $B \otimes t^\perp$ with $\dim B = h^2 - 1$, of rank

$$d_3 = (h^2 - 1)(h - 1) = 21141.$$

There are $2v_c^3 = 70317217152$ such edges. Each has an orthonormal residual basis by the retained binary-frame lemma. It is therefore applied as one child on $21141f$ selected axes, with the same sign wrappers.

The dependency-path guard. The path budget $q = m + 6h = 22120$ of Section 4.1 is unchanged. Since $2d_3 > q$, a dependency path still crosses at most one of the three selected classes. For the new class put $t = (m - d_3)/m = 811/21952$. With the same bound for $(1 - t)^{6/5}$,

$$\frac{q - d_3}{m^{6/5}} + \left(\frac{d_3}{m}\right)^{6/5} < \frac{979}{160000} + 1 - \frac{6}{5}t + \frac{3}{25} \frac{t^2}{1 - t} = \frac{372026142559}{386739360000} < \frac{999}{1000}.$$

Consequently the guard of Section 4.3 holds with three classes, with the same $\rho = 6/5$, $\bar{\theta} = 999/1000$, C_{dep} , C_0 and $C_1 = 6/5 - \beta/5 + \zeta$. The mixed-width layer of Section 5.1 has child widths f , $21896f$, $21168f$ and $21141f$.

The exact complex moment. With $B_c = 1007397164592$, $a_1 = 21896$ and $a_2 = 21168$ as in Section 6.1, the remaining singleton count is

$$S'_c = s_c - B_c(a_1 + a_2) - 2v_c^3 d_3 = 903222851311872.$$

The normalized widths are

$$r = \left(\frac{1}{21952}, \frac{391}{392}, \frac{27}{28}, \frac{21141}{21952} \right),$$

and their rank-mass weights are

$$w = \left(\frac{250477}{12693352}, \frac{12233999}{25386704}, \frac{844803}{1813336}, \frac{824499}{25386704} \right),$$

summing to $1 - \eta_c$ with $\eta_c = 5/12693352$. In addition to the logarithm bounds of Section 6.1, use $\log(21952/21141) < 37644/10^6$. For

$$a_c = \frac{18}{10^7}, \quad \sigma = 1 - a_c,$$

the same exact comparison gives

$$\Xi(1 - a_c) < \sum_{i=0}^3 \frac{w_i}{1 - a_c \ell_i} < 1,$$

with gap

$$\frac{12084702927461727916508294311187513281917509}{3098906698934144550424761963671962638378246719417509} > 3.89 \times 10^{-9}.$$

This certifies the complex saving $a_c = 18/10^7$. It exceeds the interchange saving, so $\sigma < \tau$ and the complex network does not bind.

9 Exact conditional assembly

9.1 Final multiplication parameters with source frames

Use the interchange saving $a_b = 154/10^8$ of Section 7.2 and the complex saving $a_c = 18/10^7$ of Section 8.1. The complete rational parameter choice is

$$\begin{aligned}\tau &= 1 - \frac{154}{10^8}, & \sigma &= 1 - \frac{18}{10^7}, & \epsilon &= \frac{499999}{10^6}, \\ c &= 1, & \beta &= \frac{1}{1000}, & \delta &= \frac{1}{10^{10}}, \\ \zeta &= \frac{1}{10000}, & C_1 &= \frac{11999}{10000}, \\ \lambda &= \tau + \frac{1}{10^{16}}, & \lambda' &= \tau + \frac{2}{10^{16}}, \\ \kappa &= \frac{7699}{10^{10}}.\end{aligned}$$

Since $\sigma < \tau$, the compact-control internal exponent is τ and the layer interface of the batched construction applies unchanged. Its leaf exponent is

$$\sigma + \beta(1 - \sigma) = 1 - \frac{17982}{10^{10}} < \tau.$$

The coefficient guard satisfies

$$1 - \epsilon C_1 = \frac{4000511999}{10^{10}} > 0.$$

Here $1 - 2\epsilon = 1/500000 > 0$, so the fast Gaussian hypotheses $b \geq 64d^2$, $\alpha^2\theta_i \geq 1$ and the prime-interval growth condition continue to hold, as does $K = o(\ell)$ for $K = d$. The numerical maps, their errors, the corrected chirp input enclosure and the output format are unchanged.

The seven assembly margins have the exact values

margin	value
$g_1 = 1 - \epsilon(1 + c)$	1/500000
$g_2 = \epsilon c(1 - \tau)$	38499923/500000000000000
$g_3 = \epsilon(1 - \lambda')$	3849992299500001/500000000000000000000
$g_4 = (1 - \epsilon)(1 - \tau)$	38500077/500000000000000
$g_5 = 1 - \delta - 2\epsilon$	19999/100000000000
$g_6 = 1 - \delta - \epsilon$	5000009999/100000000000
$g_7 = \epsilon$	499999/1000000

All twenty-nine parameter constraints are strict. The minimum margin is $G = g_3$, and

$$G - \kappa = \frac{492299500001}{5000000000000000000000} > 0.$$

This gap absorbs the remaining logarithmic factors. The resulting conditional multiplication bound is

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \kappa = \frac{7699}{10^{10}} = 7.699 \times 10^{-7} > 2^{-21}.$$

The exact comparisons are reproduced by `scripts/source_frame_network.py` and recorded in `certificates/source-frame-network.json`.

10 Precision correction to the retained Gaussian interface

The retained fast Gaussian construction generates $2^{-B}e^{\pi\alpha^2\beta_j^2}$ at precision P , where $B = \lceil 1.14\alpha^2 \rceil$, then shifts by B and multiplies by the disk-grid input u_j . The exponential interface gives absolute error at most 2^{1-P} before the shift. Thus the error after shifting is controlled by 2^{B+1-P} , not by a constant multiple of $2^{-P}e^{\pi\alpha^2/4}$ uniformly in α .

Set $F = 2^B$ in the retained chirped-correlation block lemma. Since $|\beta_j| \leq 1/2$ and $\pi/(4\log 2) < 1.14$, this bounds the exact input magnitude. Including the product and rounding, it also gives the required input accuracy $|\tilde{f}_j - f_j| \leq 2^{4-P}F$. The distinction matters: $2^B/e^{\pi\alpha^2/4}$ is unbounded, so the smaller exact-magnitude bound alone does not justify this accuracy.

No additional asymptotic precision is needed. For the inherited integer parameters $p > 100$ and $2 \leq \alpha < \sqrt{p}$, let $L_A = 3\lceil\sqrt{p}/(2\alpha)\rceil + 1$ and $\Gamma = 29p$. Then $L_A \leq p$, $\lceil\log_2 L_A\rceil \leq p - 1$ and $B \leq 1.14p + 1$. The block lemma requires

$$P \geq p + \Gamma + \lceil\log_2(FL_A)\rceil + 11.$$

Its right side is at most $32.14p + 11 < 34p$, so the retained $P = 34p$ satisfies the corrected error budget. The packing widths remain $O(p)$, and the target errors, tape-time exponents and final rounding guarantee are unchanged. The combined manuscript patch includes this correction directly in the retained Gaussian proof.

11 Further levers

Two further batchable classes are not used here. The stage-two data entrances have rank $(h-1)^2 = 729 > H/2$ in the fast factor V . Their pivots, after the controlled basis, lie in one row block and one column block, and they would give about 3.5% more interchange saving. In the complex network the same source frames apply, but they are not needed because $\sigma < \tau$. The interchange denominator is now dominated by the singleton climbs of the stage-one and stage-three auxiliary roles, which are set by the side producer and its gate frames.

12 Reproduction and attribution

Run `make verify` for the finite producers, exact certificates, regression checks and manuscript patches. Run `make source-frame-note` to rebuild this note. The new arguments and certificate are by eumemic, with substantial assistance from Claude (Anthropic). The batched recursion is by icekylinx with OpenAI GPT-6 Astra and Codex assistance. The finite producers are by Zhihao Chen (jacklightChen) with GPT-6 Astra assistance. Earlier constructions retain the attribution in the bibliography and NOTICE.

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